SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 7

TVM II — Effective Rates, Loans, Bonds & Stocks

The final MK2 week applies TVM to everyday manager problems: comparing bank offers (EAR), loan payments and balances (amortization), debt capacity (DSCR), terminal values via growing perpetuity, bond valuation and Gordon-model stock valuation. Cases: the Sentra Residence & Suites acquisition, and your own Week 7 pre-assignment.

Material

1 Nominal vs effective rates (EAR)

$$EAR = (1 + r/m)^m - 1; \text{ continuous: } e^r - 1$$

Equal to the nominal rate only with annual compounding; more frequent compounding raises EAR up to the continuous cap. 1.5%/month → APR 18% but EAR 19.56%. Upfront fees raise the all-in cost.

2 EAR calculator: don't be fooled by nominal rates Deep dive

Default = Bank A (9.60% monthly → 10.03%). Bank B (9.25% quarterly) → 9.58%.

3 Loans: payments, amortization, balloon, DSCR

$$Payment = P \times i \div [1 - (1+i)^{-N}]$$

Early payments are mostly interest. If maturity is shorter than amortization, a balloon must be repaid or refinanced (refinancing risk). **DSCR** = cash available

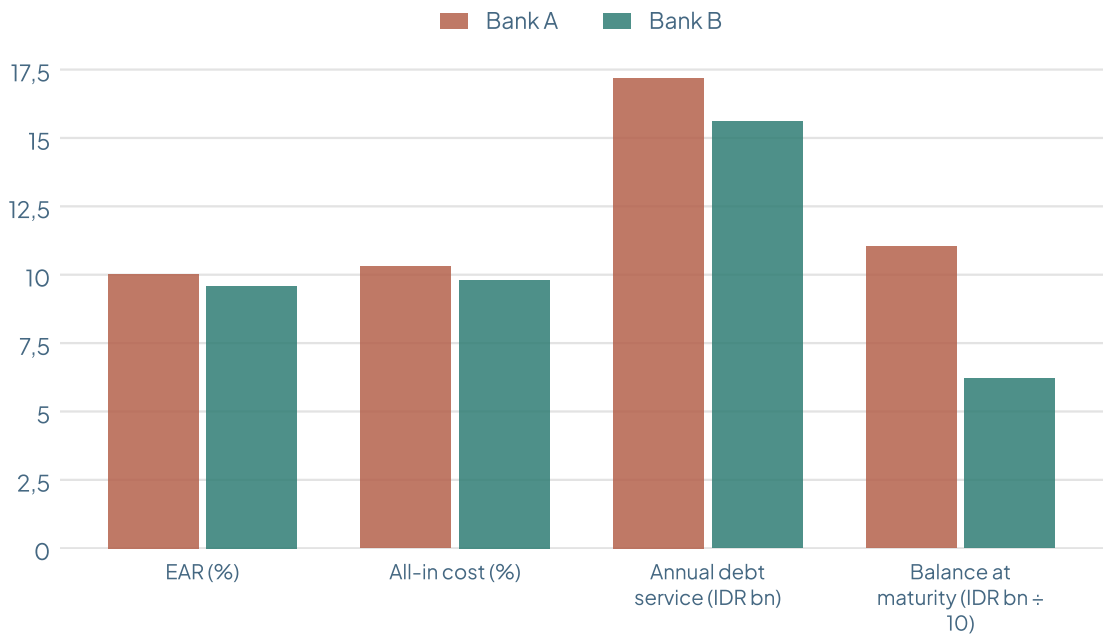
for debt service ÷ annual debt service; below 1.0× can't cover payments; lenders want ~1.2–1.3×. Exclude one-off prepayments.

4 Simulating Bank A vs Bank B loans

Deep dive

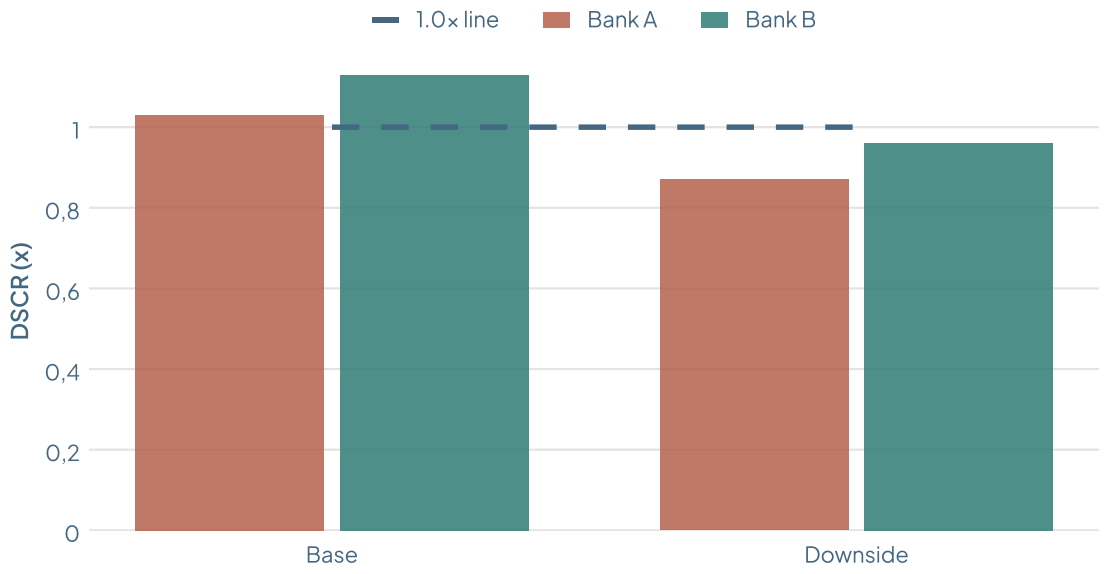
The calculator fully amortises the loan; in the case Bank A uses a longer schedule (IDR 1.434bn/month) leaving a IDR 110.3bn balloon in year 5. Bank B: IDR 126bn, 3.904bn per quarter, 10–year maturity.

Bank A vs Bank B (Sentra case)



Last row divided by 10: A 110.3bn in year 5, B 62.0bn in year 10.

DSCR: base vs downside



! DSCR below 1

Cash does not cover debt service in the downside.

5 Terminal value and the double-counting trap

Value = explicit cash flows + terminal value: $TV_{10} = CF_{11} \div (r - g)$, discounted back. TV is often the biggest and most rate-sensitive piece ($CF_{11} = 19$, $g = 3\%$: PV 133.8 at 9%, 93.3 at 10.5%, 68.0 at 12%). Don't double-count an income stream already inside TV.

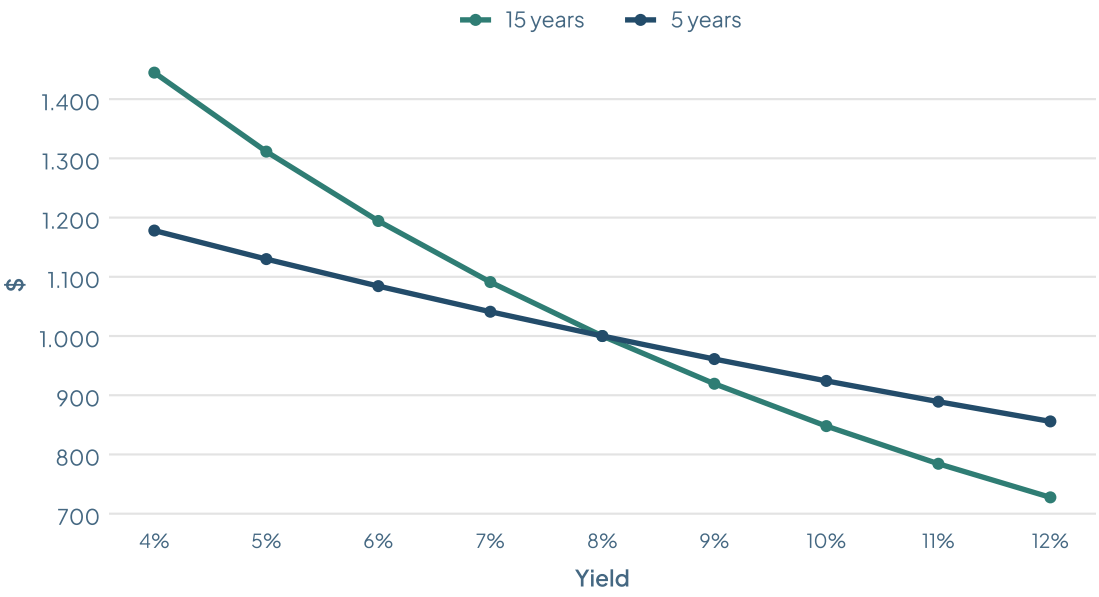
6 Bond valuation

$$B_0 = I[1 - (1+r_d)^{-n}]/r_d + M(1+r_d)^{-n}$$

r_d = coupon → par; above → discount; below → premium. Prices move opposite to rates; long bonds are more rate-sensitive. Semiannual: halve coupon and rate, double periods. Covenants and trustees protect bondholders.

7 Bond prices move opposite to interest rates Deep dive

8% coupon bond price vs yield



Yield < coupon
Premium.

Yield = coupon
Par.

Yield > coupon
Discount.

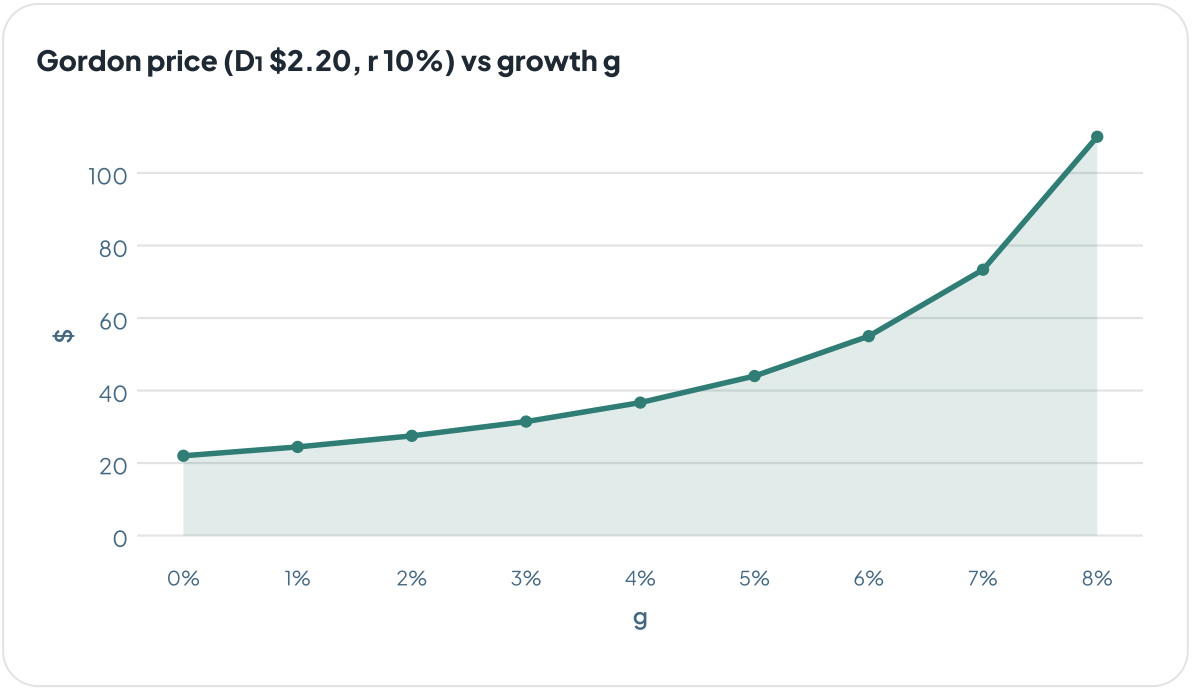
Longer bonds are more sensitive.

8 Stock valuation: dividend models

Shareholders are riskier: last in line, no maturity, no tax shield. Zero growth: $P_0 = D \div r_s$. Gordon: $P_0 = D_1 \div (r_s - g)$, $g < r_s$. Deferred dividends: value at the year before the first dividend, then discount. The gap $(r_s - g)$ drives value. Models estimate fair value; markets are largely but not perfectly efficient.

9 Gordon model: why r – g dominates Deep dive

$$P_0 = D_1 \div (r - g)$$



With D1 $2.20 and r 10%, raising g from 4% to 8% lifts price from $36.67 to $110.

Key takeaways

1. EAR = (1 + r/m)^m – 1; compare offers by EAR.
2. Loan payments are annuities; early ones are mostly interest.
3. Maturity shorter than amortization = balloon and refinancing risk.
4. DSCR = cash for debt service ÷ annual debt service; below 1× is dangerous.
5. Terminal value = next CF ÷ (r – g); avoid double counting.
6. Bond prices fall when rates rise; long bonds are more sensitive.
7. Stocks: Gordon P0 = D1 ÷ (r – g); the r – g gap drives value.

Case studies

Case & worked answers

TVM II case: Sentra Residence & Suites — acquisition and bank choice

Valuation (Rp bn): apartment annuity 72.2; retail annuity due 29.9; coworking C2 (17.5) beats C1 (13.6; break-even 3.34/yr); parking 1–10 = 7.2; terminal value 253.3 → PV 93.3. Total ≈ 220.1 vs 230 investment → ≈ (9.9). Highly sensitive to r and g; TV ≈ 42% of value.

Financing: Bank A EAR 10.03% (all-in ~10.3%), payment 1.434/month, balloon 110.3 at year 5; Bank B EAR 9.58% (~9.8%), 3.904/quarter, 101.2 at year 5, 62.0 at year-10 maturity. DSCR base 1.03× (A) vs 1.13× (B); downside 0.87× vs 0.96×.

Recommendation: prefer Bank B (cheaper, longer maturity, less refinancing risk), but coverage is thin — negotiate the price toward ~220, use less debt or reserve cash.

Case & worked answers

Your Week 7 pre-assignment (EAR, bonds, stocks) — notes and links

Your answers are complete and consistent: EAR 23.14%; Bank A 6.14% vs B 6.05%; continuous 9.42% and $3,274.91; APR 18% vs EAR 19.56%; bonds $889.30 (discount), $1,000/$1,194.24/$847.88, semiannual $887.66; stocks $50.00, $36.67, P_3 $8.33 → P_0 $5.93, $60 unchanged when r and g both rise. Good practice: you flagged the 9% vs 7% coupon inconsistency and D_0 vs D_1.

Links: EAR compares Sentra's banks; Gordon is the same formula as the property terminal value; long-dated cash flows are most rate-sensitive, as in long bonds and Garuda/Samudra's terminal values.

Worked examples

Problem 1 **Easy** **EAR: membandingkan dua tawaran**

Bank A: 12% per tahun, majemuk bulanan. Bank B: 12,4% per tahun, majemuk kuartalan. Bagi peminjam, mana yang lebih murah?

1. ```
   EAR = (1 + r/m)^m - 1
   ```
2. Bank A: $(1 + 0{,}12/12)^{12} - 1 = 1{,}01^{12} - 1 = 12{,}68\%$.
3. Bank B: $(1 + 0{,}124/4)^{4} - 1 = 1{,}031^{4} - 1 = 12{,}99\%$.

Answer: Bank A lebih murah (EAR 12,68% vs 12,99%), walaupun selisih bunga nominalnya kecil.
   ```

Problem 2 Medium **Cicilan pinjaman**

Pinjaman Rp300 juta, bunga 10% per tahun, tenor 5 tahun, dicicil bulanan. Berapa cicilan per bulan dan total bunganya?

1. i = 10% ÷ 12 = 0,8333% per bulan; n = 60 bulan.
2. `Cicilan = P × i ÷ [1 - (1 + i)^-n] = 300 × 0,008333 ÷ [1 - 1,008333^-60]`
3. $1{,}008333^{-60} = 0{,}6078$ → penyebut 0,3922 → cicilan = 2,5 ÷ 0,3922 = 6,374.
4. Total dibayar = 6,374 × 60 = 382,4 → total bunga = 82,4.

Answer: Cicilan ± Rp6,37 juta/bulan; total bunga ± Rp82,4 juta.

Problem 3 Medium **Harga obligasi**

Obligasi nominal Rp1.000, kupon 8% per tahun (dibayar tahunan), jatuh tempo 5 tahun. Yield pasar 10%. Berapa harganya?

1. Kupon = 8% × 1.000 = 80 per tahun.
2. PV kupon = $80 \times [1 - 1{,}1^{-5}] \div 0{,}1 = 80 \times 3{,}7908 = 303{,}3$.
3. PV nominal = $1.000 \div 1{,}1^{5} = 620{,}9$.
4. Harga = 303,3 + 620,9 = 924,2. Karena yield (10%) > kupon (8%), obligasi dijual di bawah nominal (diskon).

Answer: ± Rp924,2 (diskon).

Problem 4 Easy **Model Gordon**

Dividen tahun depan (D_1) Rp500, return yang diminta 12%, pertumbuhan dividen 5% selamanya. Berapa nilai wajar saham?

1. $P_0 = D_1 \div (r - g) = 500 \div (0{,}12 - 0{,}05)$
2. $= 500 \div 0{,}07 = 7.142{,}9$.

Answer: ± Rp7.143 per saham.

Problem 5 Easy **Continuous compounding**

A nominal 10% a year is compounded continuously. What is the EAR? Compare with monthly compounding.

1. $\text{EAR (continuous)} = e^r - 1 = e^{0}.10 - 1 \approx 10.517\%$
2. Monthly: $(1 + 0.10/12)^{12} - 1 \approx 10.471\%$.
3. Continuous compounding is the upper limit: the more often interest compounds, the higher the EAR, but the gains shrink.

Answer: Continuous EAR ≈ 10.52% (monthly ≈ 10.47%).

Problem 6 Medium **DSCR and the maximum debt service**

Cash flow available for debt service (CFADS) is IDR 1,200 million a year. Principal + interest payments are IDR 900 million a year. The bank requires a minimum DSCR of 1.25. Does it pass? What is the maximum payment allowed?

1. `DSCR = CFADS ÷ Total debt service = 1,200 ÷ 900 ≈ 1.33`
2. 1.33 ≥ 1.25 → **passes**, with about a 7% cushion.
3. Maximum debt service = CFADS ÷ minimum DSCR = 1,200 ÷ 1.25 = **IDR 960 million** a year.

Answer: DSCR 1.33 (passes); maximum IDR 960 million/year.

Problem 7 Challenging **Terminal value and its PV**

Explicit cash flows are projected to year 10. Year-11 cash flow is forecast at IDR 110 billion, growing 4% forever. Discount rate 11%. Compute the terminal value at year 10 and its present value.

1. $TV_{10} = CF_{11} \div (r - g) = 110 \div (0.11 - 0.04) \approx 1{,}571.4$

2. $PV = TV_{10} \div (1 + r)^{10} = 1{,}571.4 \div 1.11^{10} \approx 553.4$

3. Use the year-11 cash flow (not year 10) to avoid double counting, and discount 10 years because TV sits at the end of year 10.

Answer: TV10 ≈ IDR 1,571 billion; PV ≈ IDR 553 billion.

Practice questions

Try answering first. The answer key is below each question.

1. 12% compounded monthly. EAR ≈ …

A. 12.00%
B. 12.36%
C. 12.68%
D. 13.00%

Answer: C. 12.68%.

2. An 8% coupon bond when market yield is 10% sells at…

A. Premium
B. Par
C. Discount
D. Unsellable

Answer: C. Yield > coupon → discount.

3. D_1 $2.20, r 10%, g 4%. Gordon price…

A. $22.00
B. $36.67
C. $55.00
D. $110.00

Answer: B. 2.20 ÷ 0.06.

---

4. A DSCR of 0.87× means…

A. Cash covers debt with 87% to spare
B. Cash covers only 87% of debt service
C. 87% repaid
D. 0.87% interest

Answer: B. Below 1 = shortfall.

---

5. Why was Bank B recommended?

A. Bigger loan
B. Lower EAR/cost, lighter payments, no big year-5 balloon
C. Monthly compounding
D. DSCR above 2

Answer: B. Bank A carries refinancing risk.

Sources

- MBA_TVM_II_Excel_Companion_Student.xlsx
- 130_MK 2 Week 7 (pre-assignment: Effective Rates, Bond & Stock Valuation)